Electronics For You

Vol. 42 No. 7 JULY 2010

42nd Year of Publication

Since January 1969

www.efymag.com

ISSN-0013-516X

EDITOR : Ramesh Chopra

EDITORIAL CORRESPONDENCE : Editorial Secretary
Phone: 011-26810601
E-mail: editor@efyindia.com
(Technical queries: efyfaq@efyindia.com)

SUBSCRIPTIONS & MISSING ISSUES : Customer Care Officer, EFY
Phone: 011-26810601 or 02 or 03
E-mail: support@efyindia.com

BACK ISSUES, BOOKS, CDs, PCBs etc. : Ritu N Tijauria,
D-88/5, Okhla Industrial Area, Phase 1
New Delhi 110020
Phone: 011-2637 1661, 26371662
E-mail: info@ritutijauria.com

EXCLUSIVE NEWSSTAND DISTRIBUTOR : Bhl Books & Magazines Distributors Pvt Ltd
Ach.No. 30, Below Mahalaxmi Bridge
Mahalaxmi, Mumbai-400034
Phone: 022-40497401, 40497402, 40497474, 40497413; Fax: 40497434
E-mail: circulations@bhlworld.com

ADVERTISEMENTS NEW DELHI (HEAD OF OFFICE) : Poonam Sharma
D-8/71, Okhla Industrial Area, Phase 1,
New Delhi 110020 INDIA
Phone: 011-26810601 or 02 or 03 Fax: 26817561
E-mail: efyenq@efyindia.com

MUMBAI : Vijay D'Souza
307, Shivsmruti Chambers, 4th Dr A.B. Road, Worli Naka
Mumbai 400018; Ph: 022-24950847, 24928520
Fax: 24954229; E-mail: efymum@efyindia.com

BENGALURU : Jayashree
No. 9, 17th Main, 1st Cross, HAL II Stage
Indiranagar, Bengaluru 560008
Ph: 080-25260194, 25260023; Fax: 25260394
E-mail: efyblr@efyindia.com

CHENNAI : DBS House, 31-A, Cathedral Garden Road
Near Palmgrove Hotel, Chennai 600034
Phone: 011-26810601 or 02 or 03 Fax: 011-26817563
E-mail: efyenq@efyindia.com

KOLKATA : D.C. Mehra
67-A, Park Street, Kolkata 700016
Ph: 033-22294760, Telefax: 22650094
E-mail: efycal@efyindia.com; Mobile: 09432422912

PUNE : Phone: 011-26810601 or 02 or 03 Fax: 011-26817563
E-mail: efyenq@efyindia.com

HYDERABAD : Ashish Kotak; Mobile: 09703646567
E-mail: efyhyd@efyindia.com

GUJARAT : Sunil Singh; Mobile: 09999661298
E-mail: efyahd@efyindia.com
Shambhukhilal Patel; Mobile: 09825210453
Ph: 079-22159840; Telefax: 0779-2213841

SINGAPORE : Ms Peggy Thay, Publicitas Singapore Pte Ltd
72, Bendemeer Road, #02-2D,
The Luzerne, Singapore 339941
Phone: +65-6856 2272 Fax: +65-6297 7302
E-mail: pthay@publicitas.com, singapore@publicitas.com

UNITED STATES : Ms Veronique Lamarque, E & Tech Media
Phone: +1 860 536 6677
E-mail: veroniquelamarque@gmail.com

CHINA : Ms Terry Qin, Power Pioneer Group Inc.
Room 419 , Zhongjian Building, No. 452
Dong Sangda Industry Park, No. 52 Zhenhua Road
Futian District, Shenzhen China 518031
Phone: (86 755) 83729797; Fax: (86 21) 6455 2579
Mobile: (86) 13923922595, 18650155918
E-mail: terryqin@powerpioneergroup.com,
gpjterryq@gmail.com

TAIWAN : Mr Jean Chen, J.K. Media
4F-7, No. 348, Sec. 2, Bade Rd., Songshan Dist.,
Taipei City 105, Taiwan
Phone: 886-2-87726780 ext 10; Fax: 886-2-87726787

Printed, published and owned by Ramesh Chopra. Printed at Nutech Photolithographers, D-30, Okhla Industrial Area, Phase-I, New Delhi, on the first day of each month and published from D-87/1, Okhla Industrial Area, Phase-I, New Delhi 110020. Copyright 2010. All rights reserved throughout the world. Reproduction of any material from this magazine in any manner without the written permission of the publisher is prohibited. Although every effort is made to ensure accuracy, no responsibility whatsoever is taken for any loss due to publishing errors. Articles that cannot be used are returned to the authors if accompanied by a self-addressed and sufficiently stamped envelope, but no responsibility is taken for any loss or delay in returning the material. EFY will not be responsible for any wrong claims made by or advertisements published in it. Any dispute will be settled in a New Delhi court only.

SUBSCRIPTION RATES

Period	Newstand Price (Rs)	You Pay (Rs)	Overseas
Five	3,000	1,800	—
Three	1,800	1,260	—
One	600	480	US$ 100

For receiving both Summer and Winter editions of EFY Directories with Yellow Pages in each priced at Rs 1000, please add Rs 500 for each year (including handling and forwarding).
Kindly add Rs 50/- for outside Delhi cheques.
Please send payments only in favour of **EFY Enterprises Pvt Ltd**
Non-receipt of copies may be reported to support@efyindia.com mentioning your subscription number

CONTENTS

30 **Technology Focus**
Telemedicine: Ophthalmologist in Theni, Patient in Aandipatti

38 **Healthcare**
Brain-Computer Interface

46 **EFY Report**
Solar PV Cells Target Grid Parity

54 **Solar**
Power From 'Solar Grid'

56 **Do-It-Yourself**
IR Protocol Analyser

62 **Embedded**
Hardware-Software Co-Design for Embedded Systems

70 **Construction**
Remote-Controlled 6-Camera CCTV Switcher

93 **Construction**
Christmas Decoration Light

102 **Software Section**
JavaScript Based Command Line Interpreter

105 **Test & Measurement**
HIPOT Testers For Electrical Safety

109 **Sensors**
Taste Sensors' Working and Applications

111 **Satcom**
Evolution of Satellite Communication

117 **Innovation**
Aerial Eye to Guard Against Terrorism

119 **IT Zone**
Computing Takes Quantum Leap

125 **Design**
Low-Power MCU Selection—Part 2 of 2

157 **Interview**
"Thin-film solar cells offer a lower-cost solution"—Mano Manoharan, general manager, GE Global Research, Bengaluru

Cover Photocourtesy: Cisco HealthPresence

Departments

EFY-CD 10
CD THEME 12
WEBSITES 14
LETTERS 16
Qs&As 18
TECHNOLOGY NEWS 22
CIRCUIT IDEAS 96
NEW PRODUCTS 130
INDUSTRY NEWS 133
BIZ ADS 138
ELECTRONICS MART 149
ELECTROONS 160
SUBJECT & ORGANISATION INDEX 162
ADVERTISERS' INDEX 163

e-Style

78 **First Look!**
- Noise-cancelling Headphone
- HD Media Player
- 12MP HD Camera
- Ultra-portable Netbook
- HD Set-top Box
- Water-resistant Walkman
- Home Theatre Systems
- Webcam with VGA Video

84 **Buyers' Guide**
Robotic Kits: Popularity on Rise

91 **Interview**
"Consumer trend is towards having multiple cards"—Manisha Sood, country manager for SanDisk India and SAARC

NEXT ISSUE

TECHNOLOGY FOCUS
Smart Energy Distribution

BUYERS' GUIDE
3D Gaming For You

SURVEY/EFY REPORT
Medical Electronics